XXXVIII JORNADAS DE INGENIERÍA QUÍMICA
Granada 9-11 julio 2025

El papel de las biorrefinerías en el reto energético y la neutralidad climática

Ángel Galán-Martín, Déborah Pérez-Almada, María del Mar Contreras, Juan Miguel Romero-García y Eulogio Castro

Universidad de Jaén

INSTITUTO DE INVESTIGACIÓN EN BIORREFINERÍAS

MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES
Cofinanciado por la Unión Europea
AGENCIA ESTATAL DE INVESTIGACIÓN

***Acknowledgement**: Project PID2023-151855OA-I00 funded by MICIU/AEI /10.13039/501100011033 y por FEDER, UE*

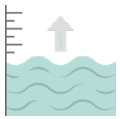
Different climate futures

Most scenarios aligned with the **Paris goal** require the large-scale deployment of **Carbon Dioxide Removal (CDR)** for two main reasons:

- To compensate emissions that very likely will exceed the carbon budget.
- To offset emissions from hard-to-abate sectors.

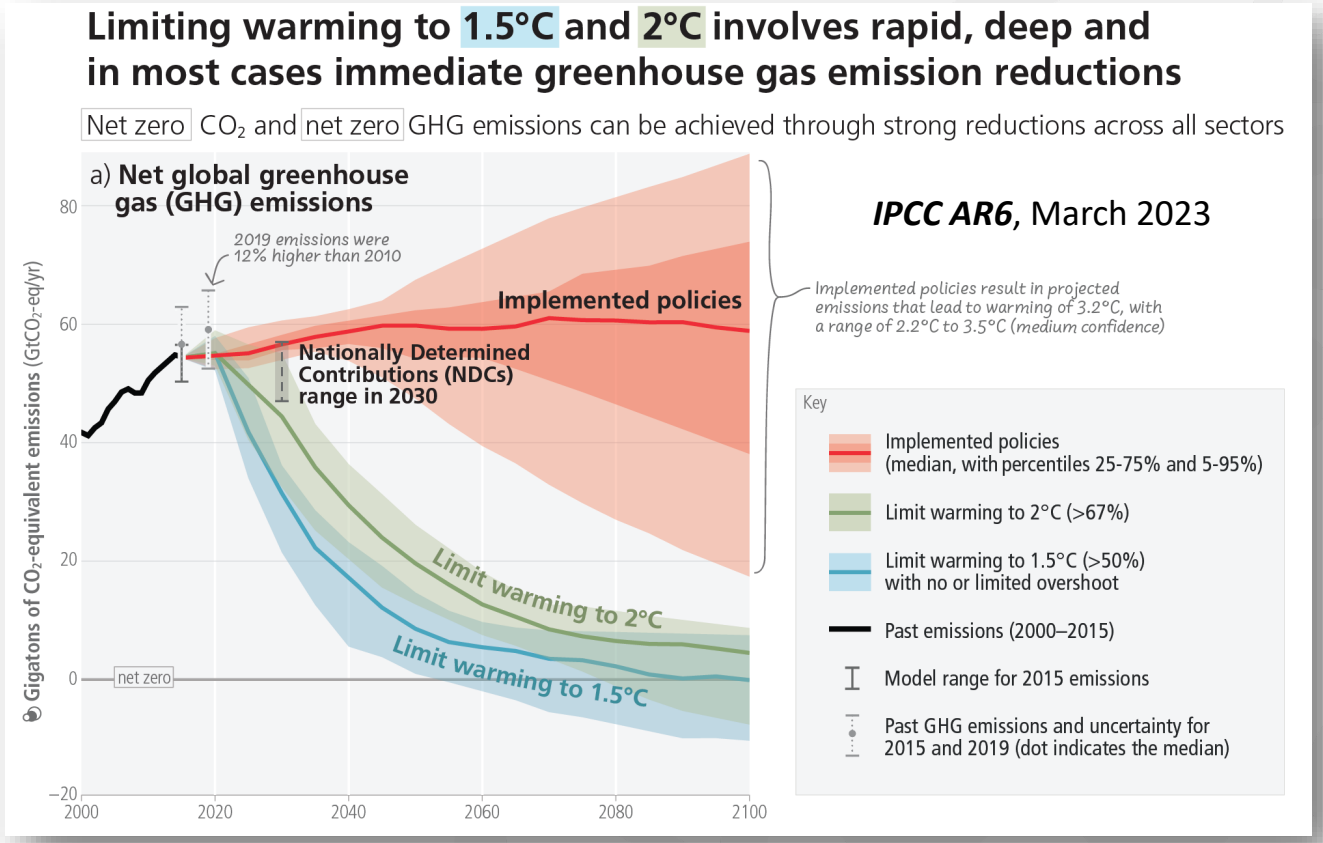








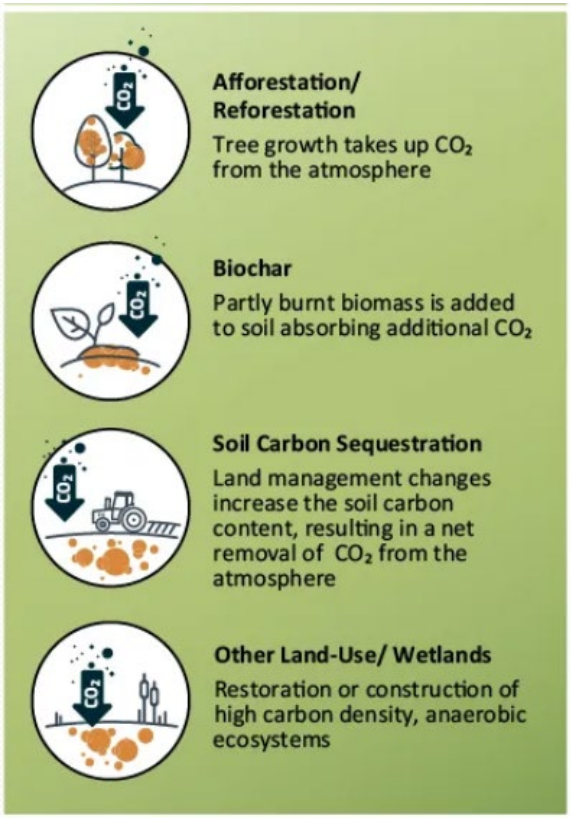


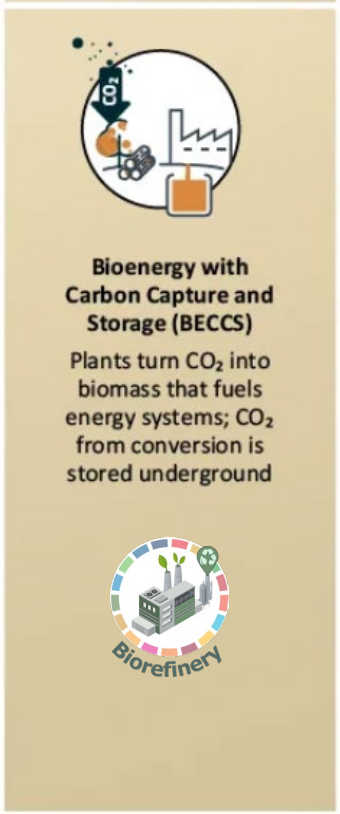


How can we remove CO_2?

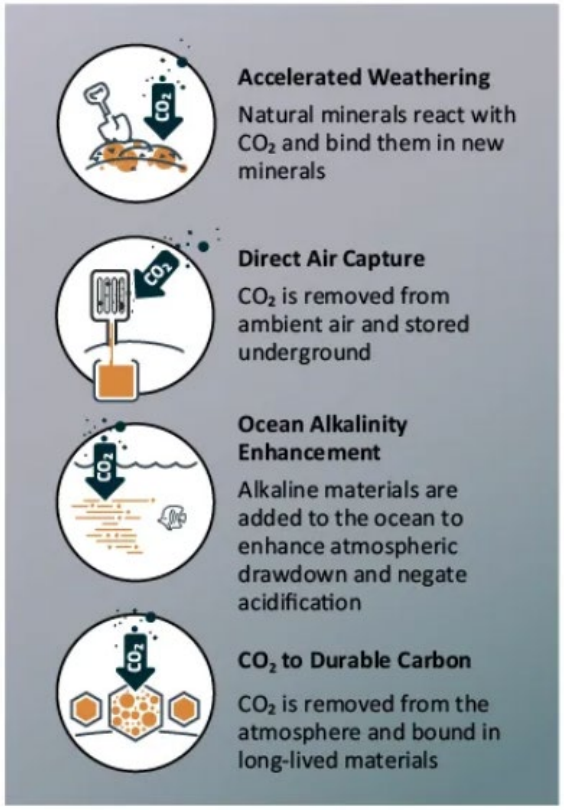
There are different strategies to deliver **CDR** but only a few of them are **realistic** and **scalable** including **natural**-based solutions and **technological approaches**.

Natural





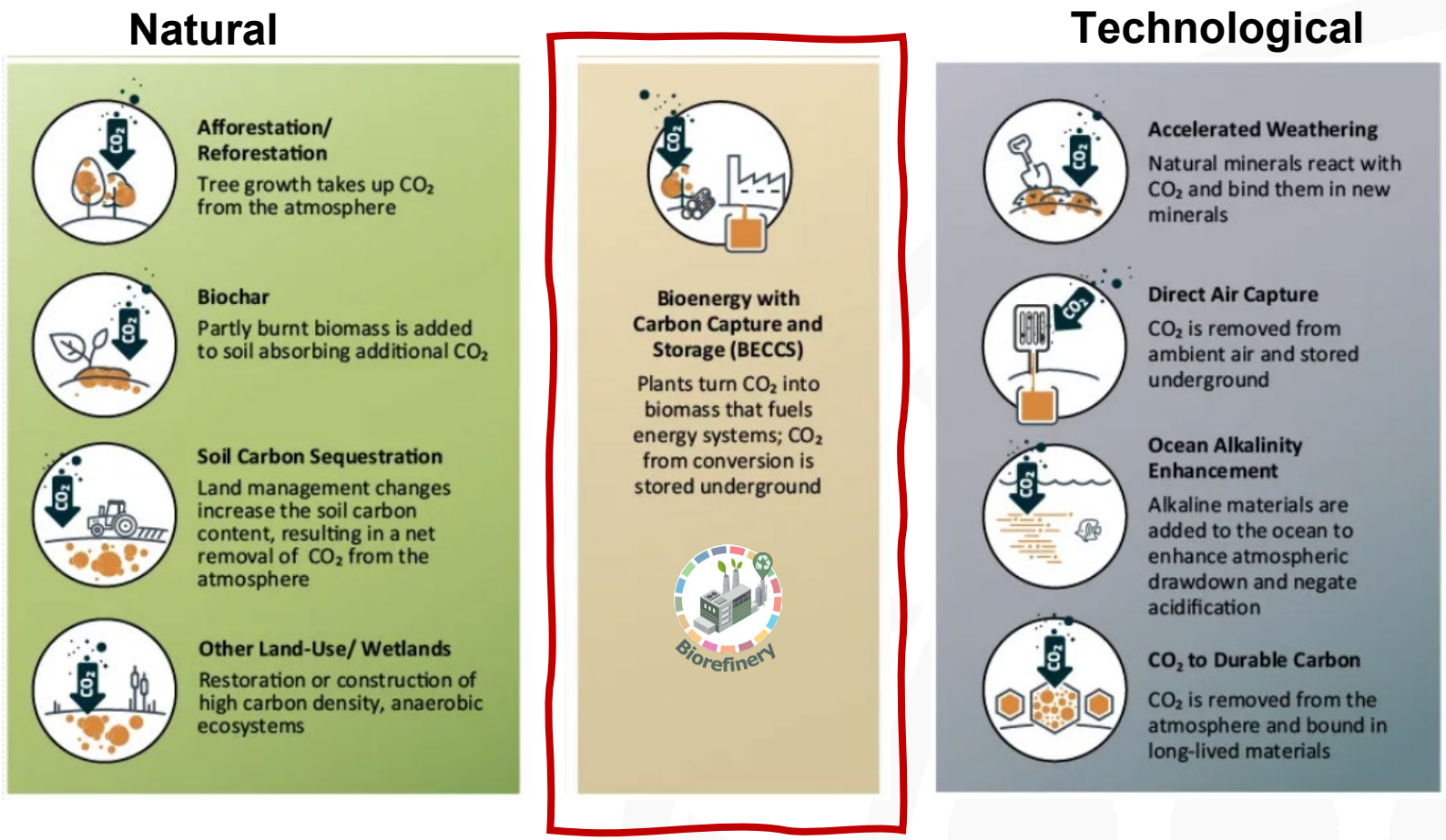
Technological



Emission Report. 2017, **UNEP**.

How can we remove CO_2?

There are different strategies to deliver **CDR** but only a few of them are **realistic** and **scalable** including **natural**-based solutions and **technological approaches**.



Emission Report. 2017, **UNEP**.

Understanding the BECCS concept

Biorefineries are industrial facilities dedicated to the conversion of biomass resources into energy and products with multiple applications.

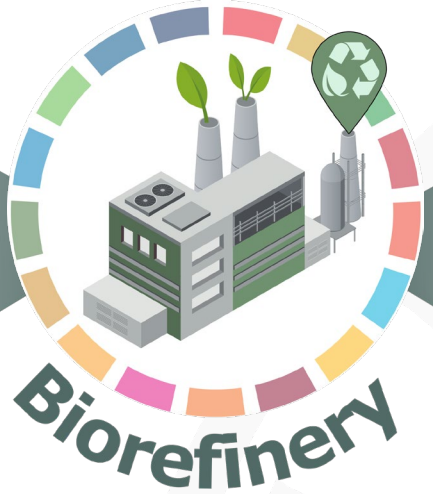


Understanding the BECCS concept

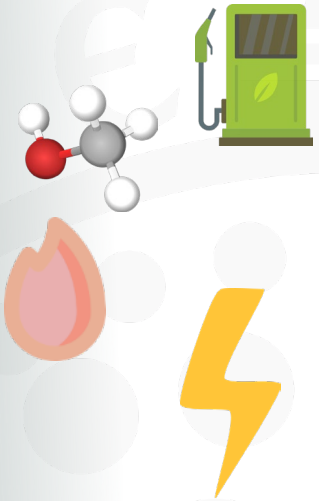
Biorefineries are industrial facilities dedicated to the conversion of biomass resources into energy and products with multiple applications.



Forestry residues
Agricultural residues
Energy crops
Municipal solid waste
…



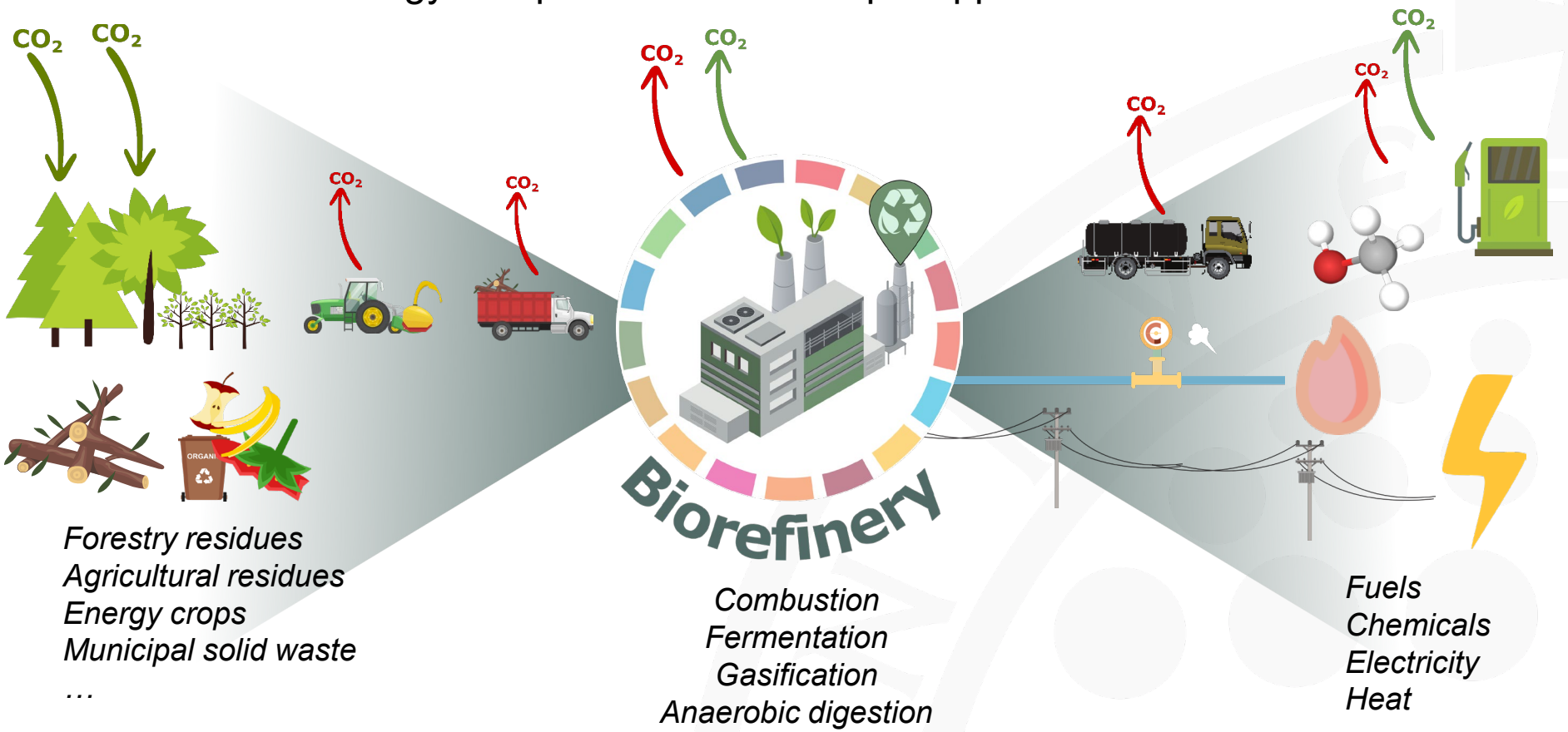
Combustion
Fermentation
Gasification
Anaerobic digestion



Fuels
Chemicals
Electricity
Heat

Understanding the BECCS concept

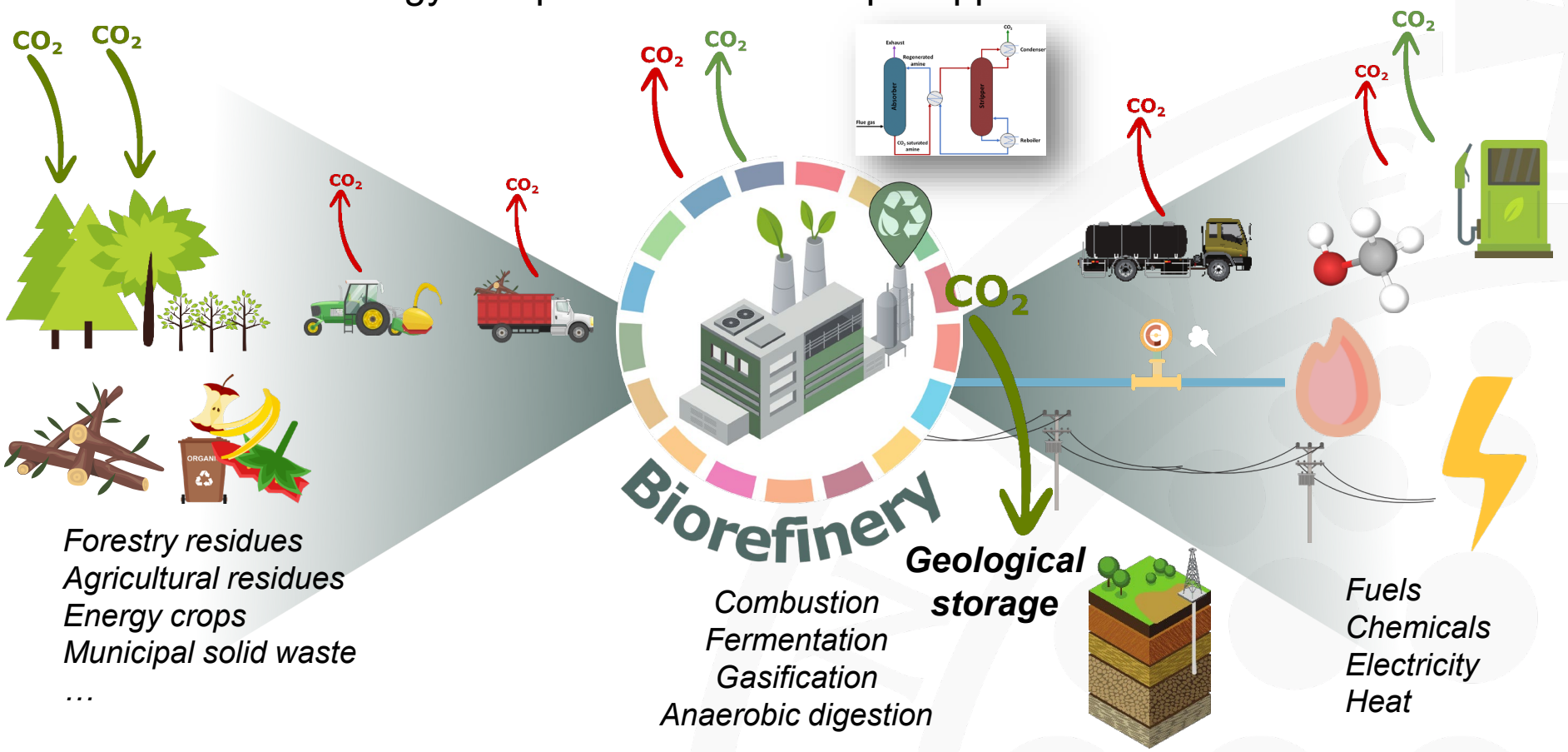
Biorefineries are industrial facilities dedicated to the conversion of biomass resources into energy and products with multiple applications.



Understanding the BECCS concept

Appl. Energy., 2020, **279**, 115884
J. Clean. Prod., 2024, **456**, 142361
Sustain. Prod. Consum., 2024, **47**, 516-527

Biorefineries are industrial facilities dedicated to the conversion of biomass resources into energy and products with multiple applications.



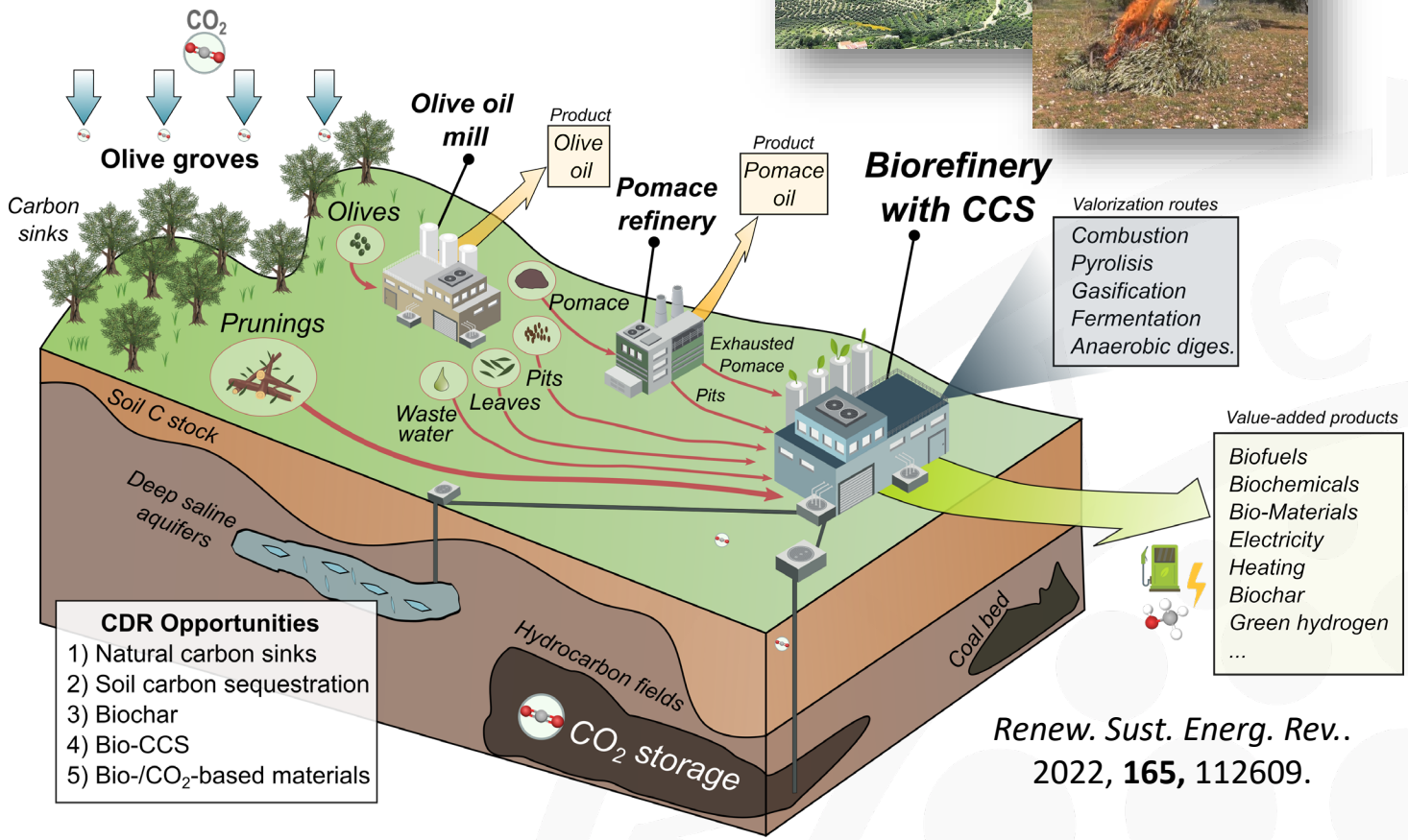
Biorefinery + CO_2 capture and storage

Carbon-negative?

Olive-derived biomasses





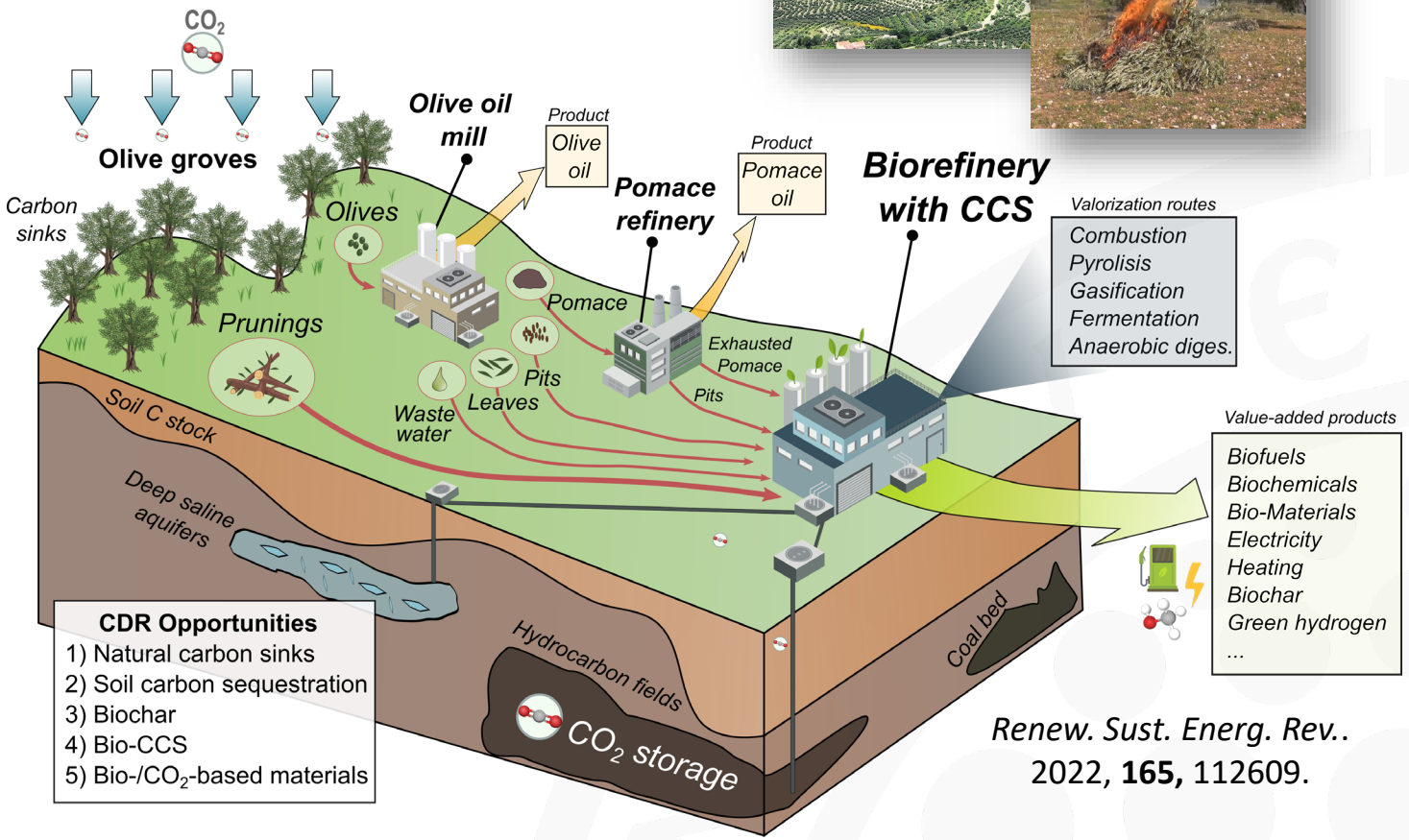


Renew. Sust. Energ. Rev.. 2022, **165,** 112609.

Olive-derived biomasses





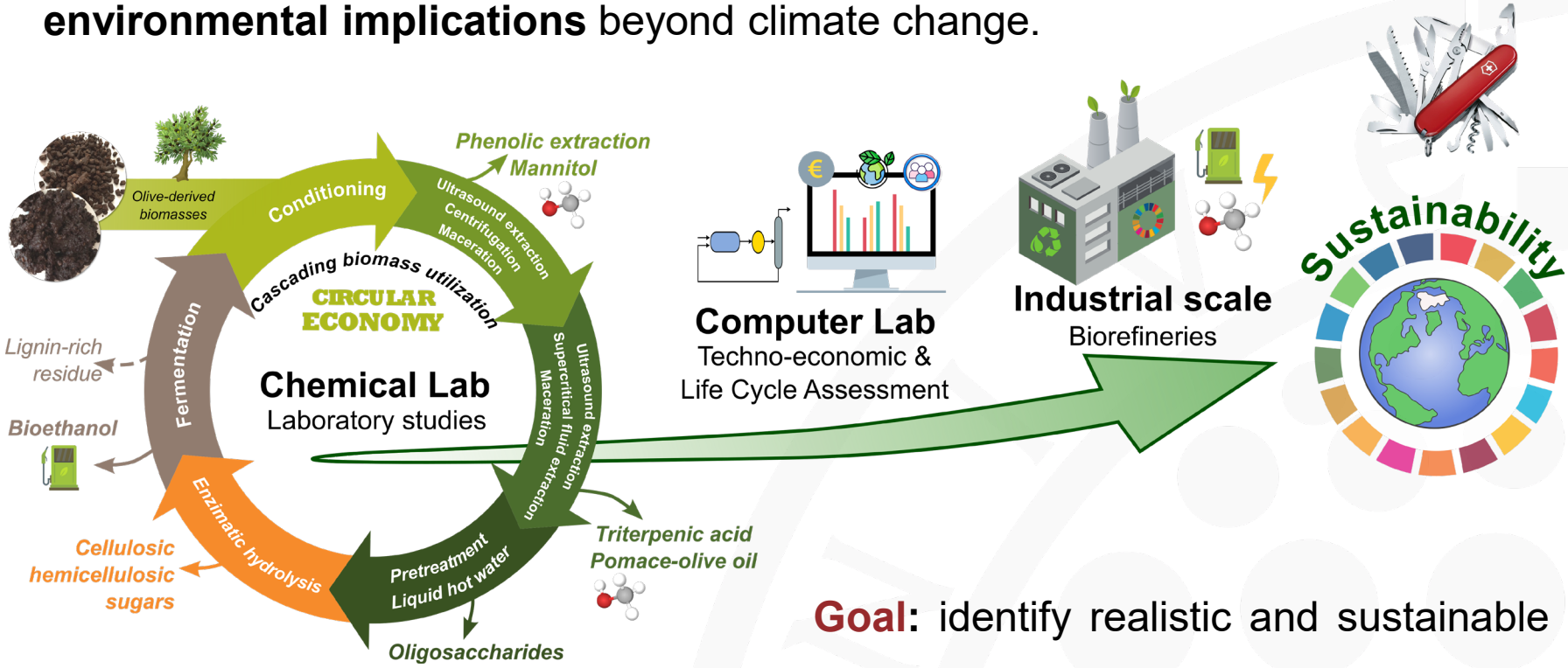


Renew. Sust. Energ. Rev.. 2022, **165,** 112609.

Case study: how integrating biorefineries w/wo CCS can valorize **exhausted olive pomace** — turning a local agro-industrial strategic residue into valuable bio-based products and energy.

Research methodology

Two levels of research: **experimental research** and industrial-scale **modeling/simulation exercises** to analyze the **technical feasibility and environmental implications** beyond climate change.

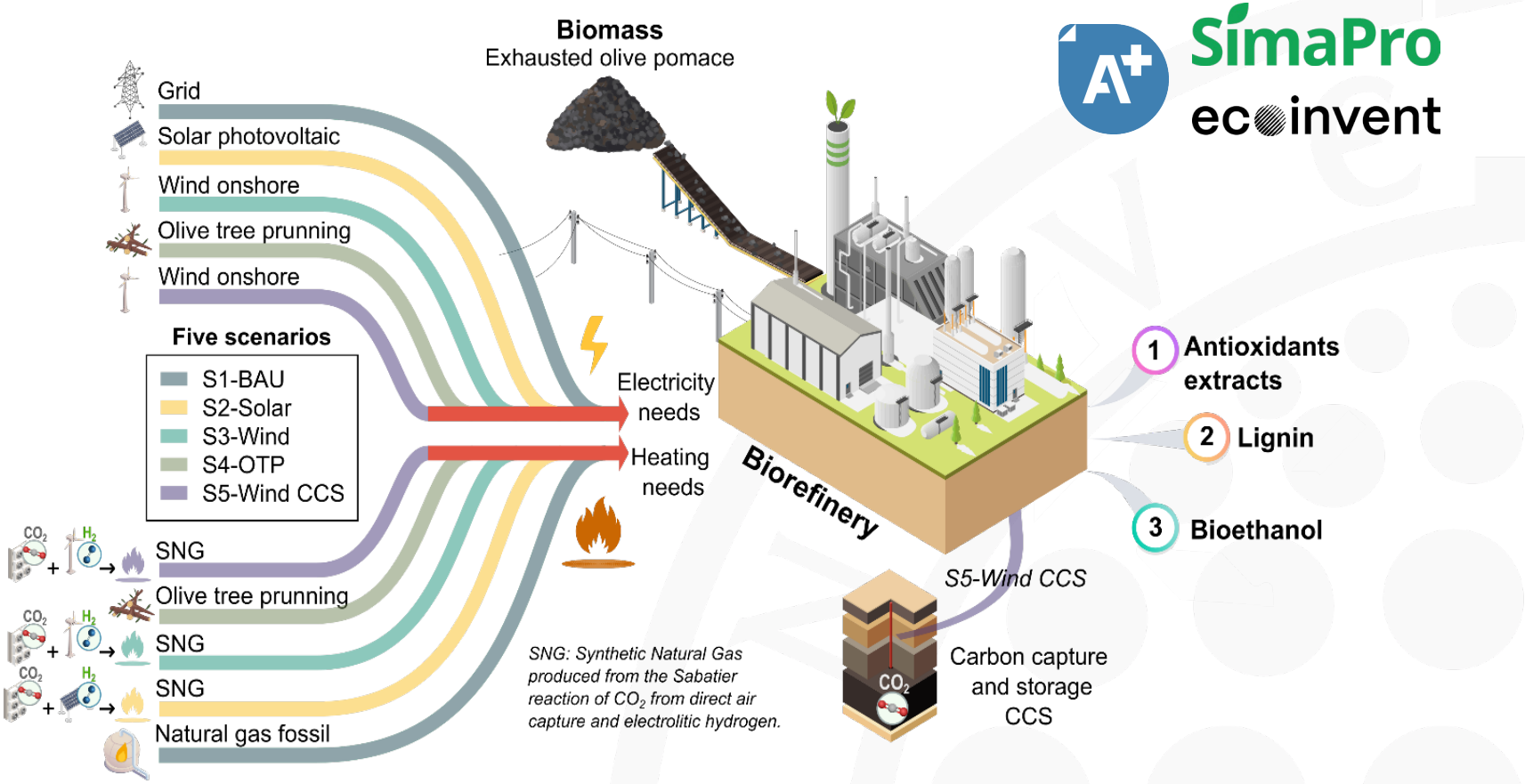


Goal: identify realistic and sustainable pathways to valorize olive pomace.

SUSPROLIBE – Sustainable Processes to valorize olive-derived biomasses towards a circular bioeconomy (TED2021-132614A-I00)

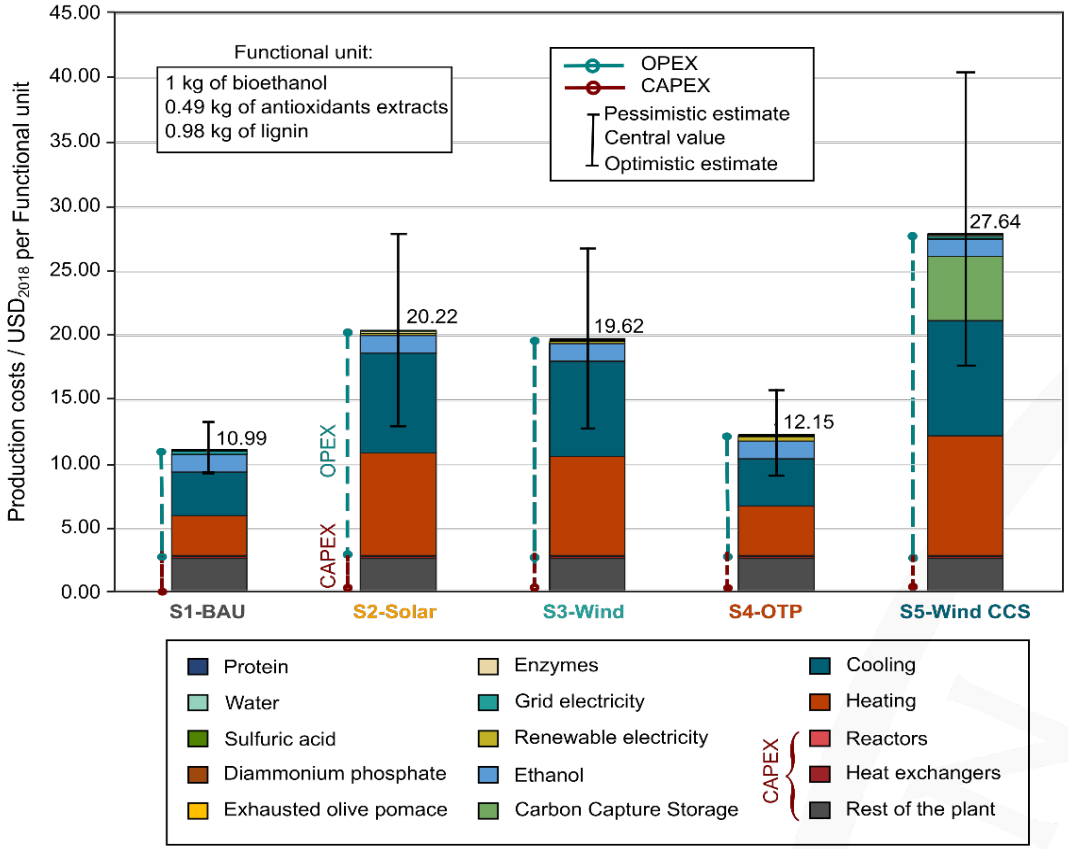
Biorefinery scenarios

Assess **techno-economic** and **enviromental implications** of **multiproduct biorefinery** system producing **antioxidant** extracts, **lignin**, and **bioethanol** from exhausted olive pomace.



ACS Sustainable Chem. Eng. 2025, **13**, 7, 2732–2745

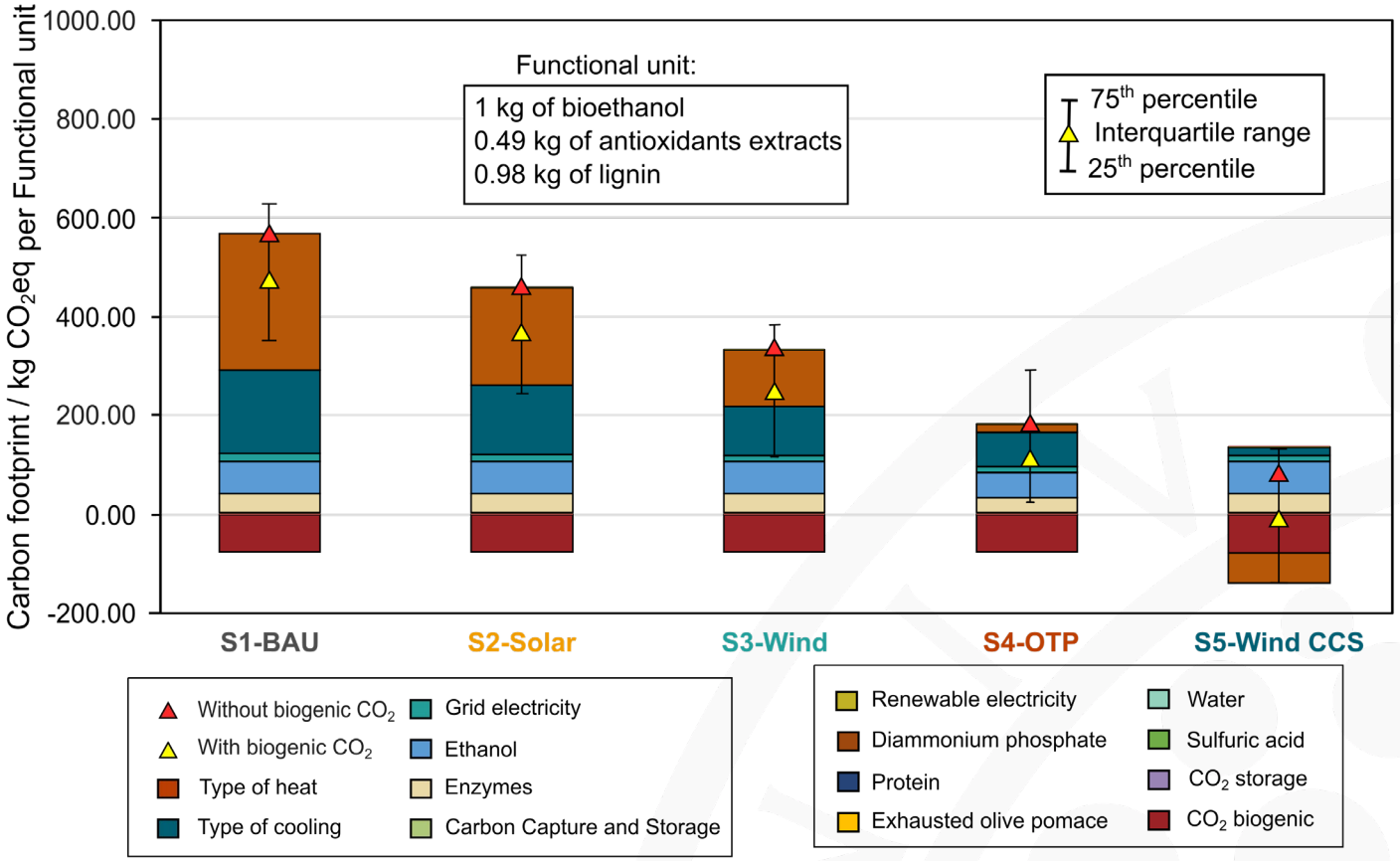
Economic assessment



- ✓ Energy needs are main contributors to the costs.
- ✓ **S1-BAU** presents the lowest cost.
- ✓ **S5-Wind CCS** presents the higher costs due to CCS infrastructure and operation.
- ✓ **S4-OTP** presents a competitive costs compared with S1-BAU.

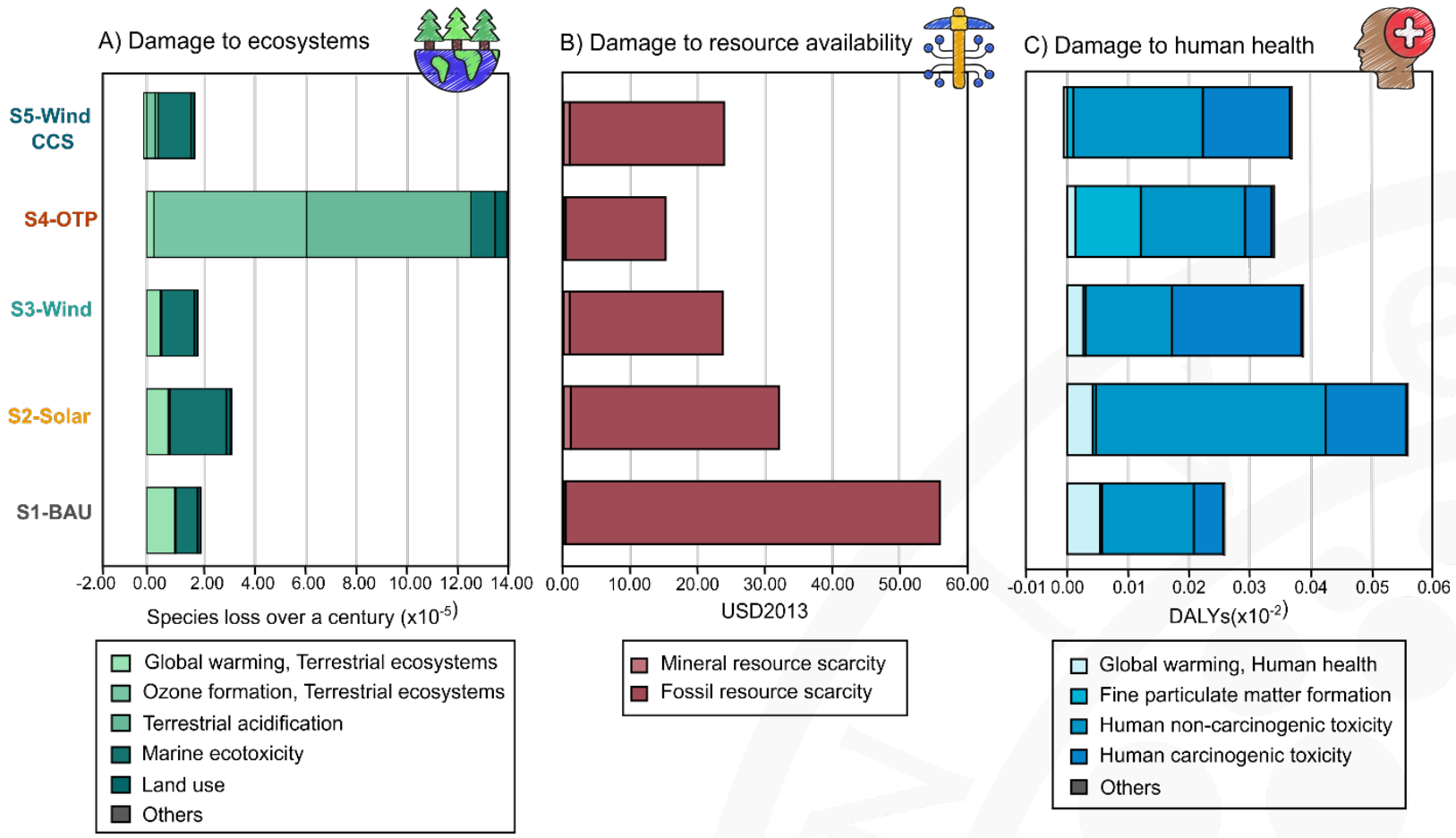
Carbon footprint

GWP100 (ReCiPe 2016)



- ✓ Heating and cooling demands are identified as the main contributors.
- ✓ **S1-BAU** shows the highest carbon footprint.
- ✓ **S5-Wind CCS** achieves a negative carbon footprint (*cradle-to-gate*).
- ✓ **Selling carbon credits** offer a promising strategy to offset extra costs.

Other environmental implications



✓ Our results reveal that there may be **trade-offs to other environmental categories**, causing increased impact on **ecosystem quality**, **resource availability**, and **human health**.

Conclusions

Unlocking the potential of biomass and biorefineries w/wo CCS is key to achieving climate and sustainability goals.

Its **multifunctional** nature offers a **strategic advantage** by integrating the production of bioenergy and bioproducts with CO_2 capture and utilization.

PSE tools and LCA are key tools to evaluating potential environmental impacts and avoiding unintended collateral damage.

The time to act is **now**.

Conclusions

Unlocking the potential of biomass and biorefineries w/wo CCS is key to achieving climate and sustainability goals.

Its **multifunctional** nature offers a **strategic advantage** by integrating the production of bioenergy and bioproducts with CO_2 capture and utilization.

PSE tools and LCA are key tools to evaluating potential environmental impacts and avoiding unintended collateral damage.

The time to act is **now**.

Conclusions

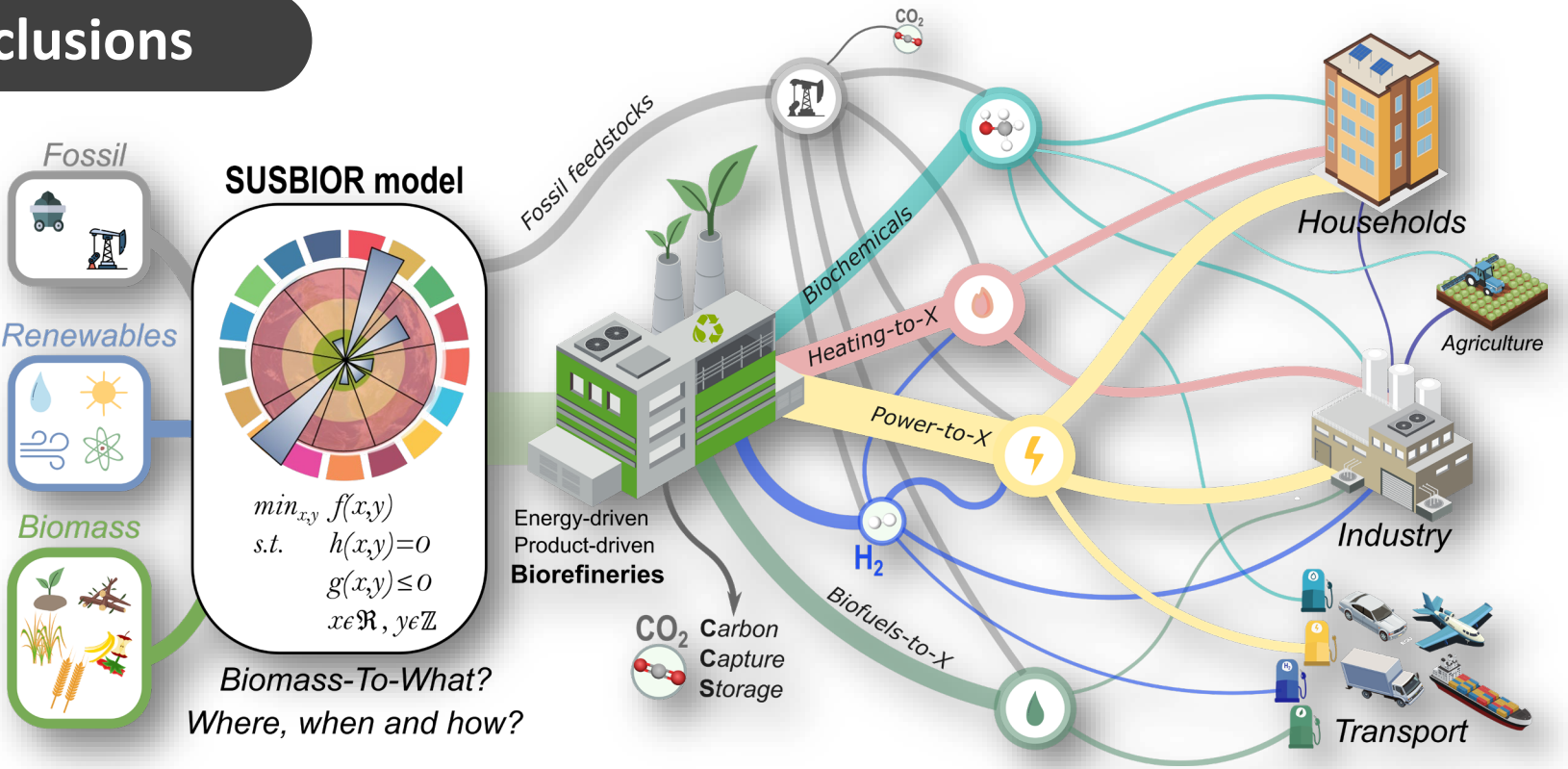
Unlocking the potential of biomass and biorefineries w/wo CCS is key to achieving climate and sustainability goals.

Its **multifunctional** nature offers a **strategic advantage** by integrating the production of bioenergy and bioproducts with CO_2 capture and utilization.

PSE tools and LCA are key tools to evaluating potential environmental impacts and avoiding unintended collateral damage.

The time to act is **now**.

Conclusions



PSE tools and LCA are key tools to evaluating potential environmental impacts and avoiding unintended collateral damage.

The time to act is **now**.

SUSBIOR *- Optimized and* ***SUS****tainable* ***BIOR****efinery supply chains in Spain for a climate-neutral European economy (PID2023-151855OA-I00)*

MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES
Cofinanciado por la Unión Europea
AGENCIA ESTATAL DE INVESTIGACIÓN

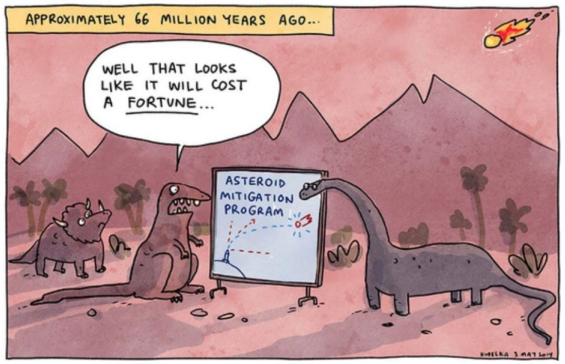
Conclusions

Unlocking the potential of biomass and biorefineries w/wo CCS is key to achieving climate and sustainability goals.

Its **multifunctional** nature offers a **strategic advantage** by integrating the production of bioenergy and bioproducts with CO_2 capture and utilization.

PSE tools and LCA are key tools to evaluating potential environmental impacts and avoiding unintended collateral damage.

The time to act is **now**.



Thank you!

Ángel Galán Martín

galan@ujaen.es

in/angelgalanmartin

@an_gel_GM

Universidad de Jaén

INSTITUTO DE INVESTIGACIÓN EN BIORREFINERÍAS

MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES

Cofinanciado por la Unión Europea

AGENCIA ESTATAL DE INVESTIGACIÓN

***Acknowledgement**: Project PID2023-151855OA-I00 funded by MICIU/AEI /10.13039/501100011033 y por FEDER, UE*